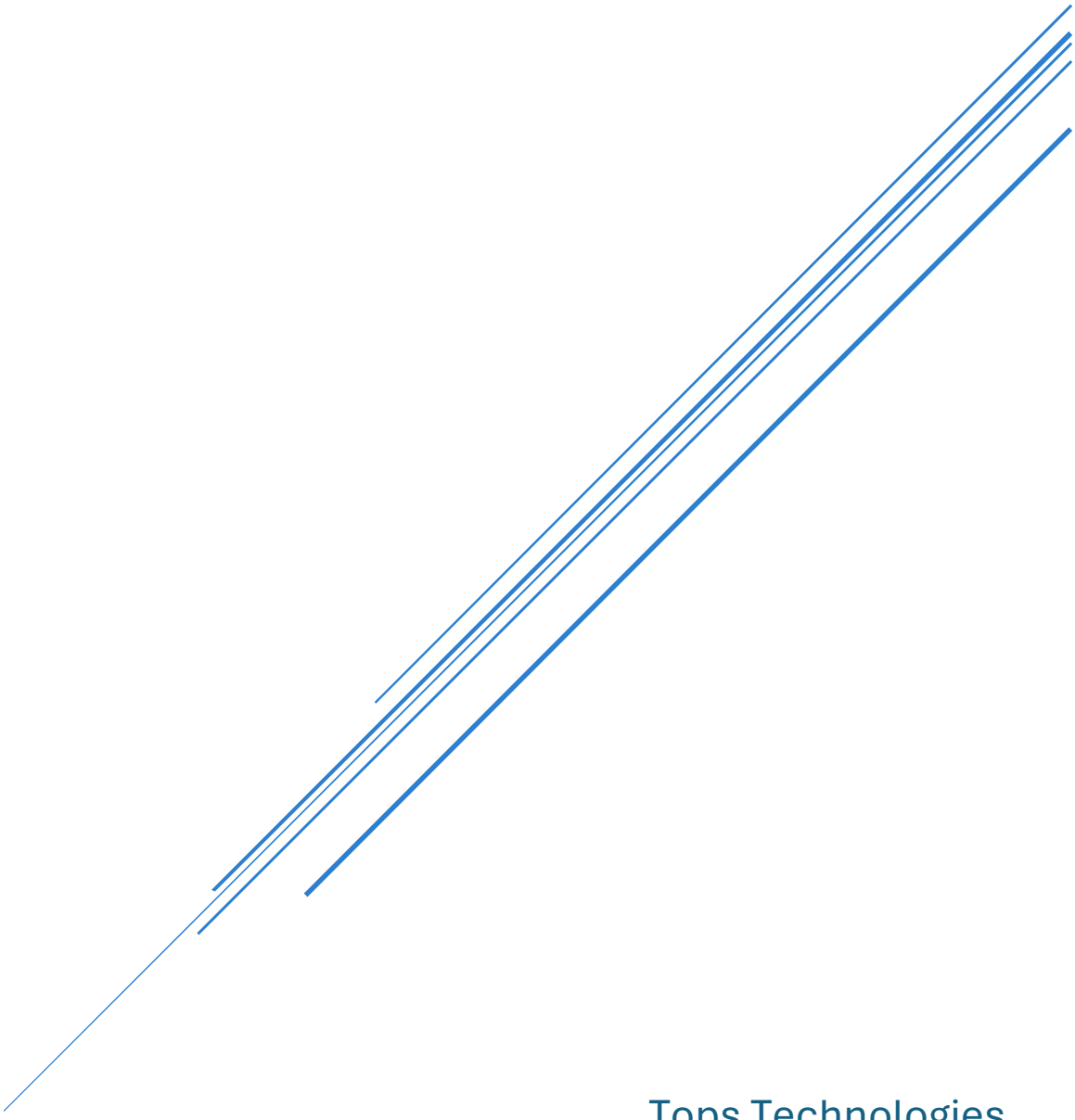
DAY – 4 FIREWALL BASICS

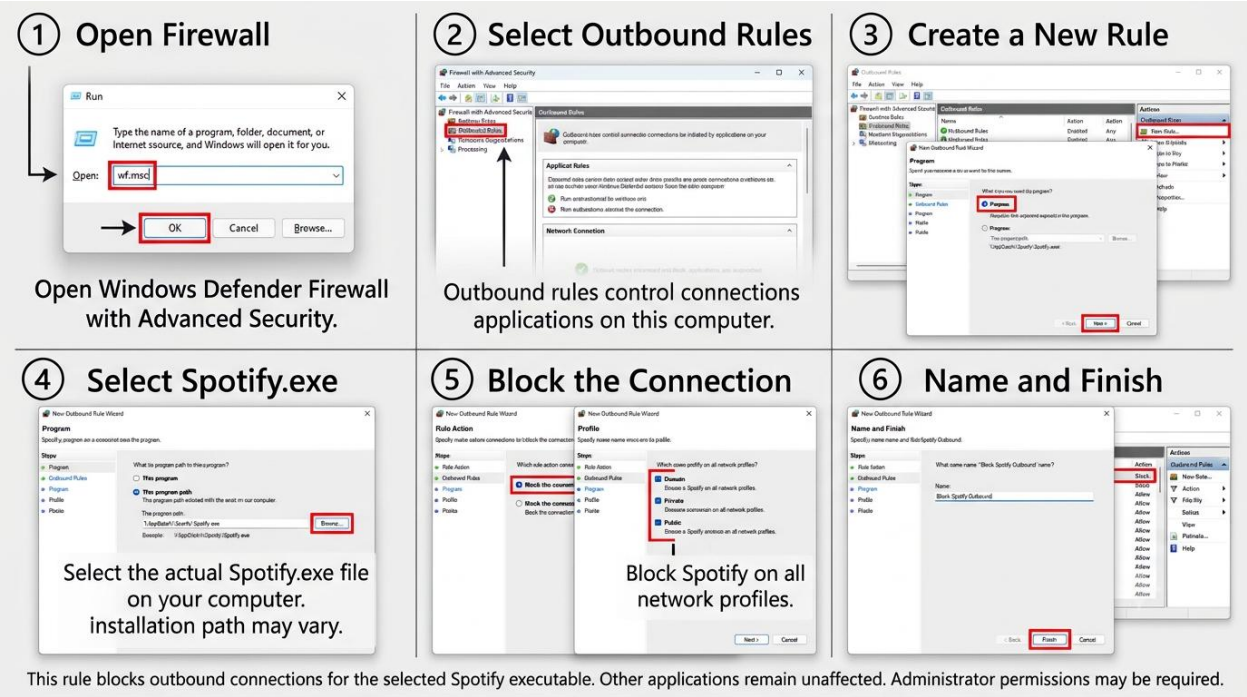
Assignment - 20



Tops Technologies
Cyber Security

1. **Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.**

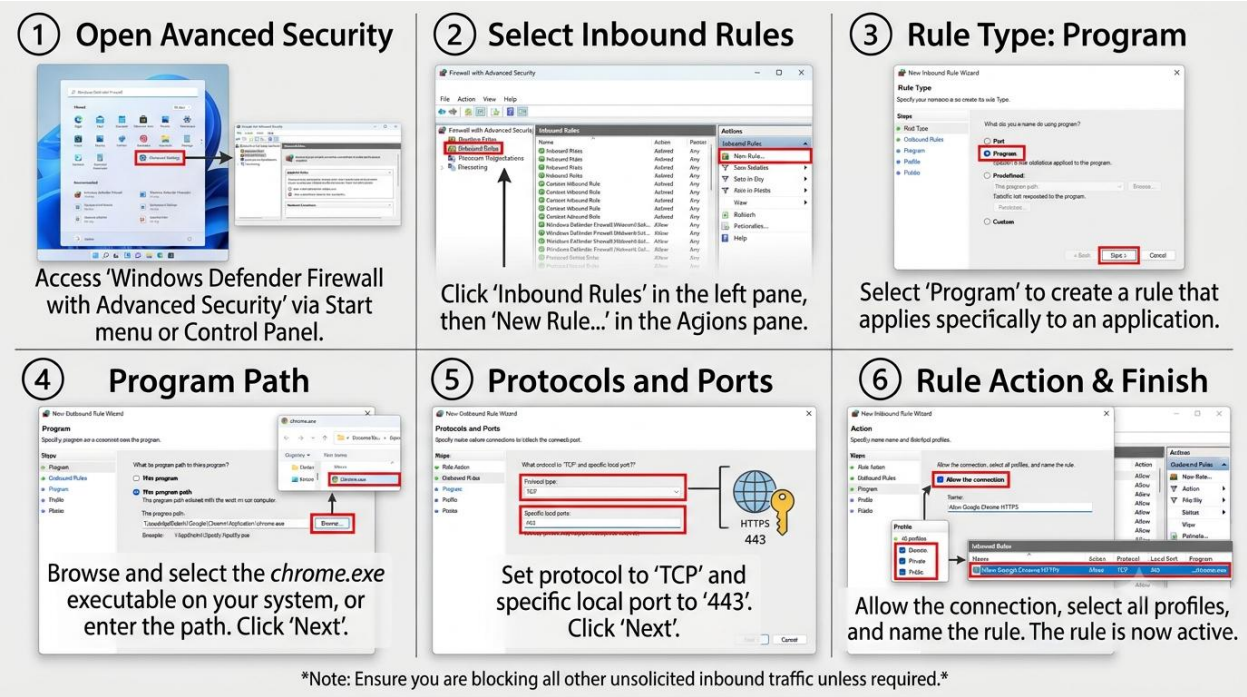
➢



This rule blocks outbound connections for the selected Spotify executable. Other applications remain unaffected. Administrator permissions may be required.

2. **Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).**

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3. **Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.**
   Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

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1. Open Windows Defender Firewall: Press Win + R, type wf.msc, and press Enter.

2. Create a new rule: In the left panel, select Inbound Rules. In the right panel, click New Rule....

3. Choose the rule type: Select Custom and click Next. This allows you to specify an IP address using the Scope tab.

4. Configure the program and protocol: Select All programs and click Next. On the Protocol and Ports screen, select Any for the protocol.

5. Specify the IP address: Open the Scope tab. Under Which remote IP addresses does this rule apply to?, select These IP addresses, click Add, enter 123.45.67.89, and click OK.

6. Block the connection: Click Next, select Block the connection, and click Next.

7. Select profiles: Check Domain, Private, and Public, then click Next.

8. Name the rule: Enter Block IP 123.45.67.89 - Inbound and click Finish.

9. Create an outbound rule: Repeat the same process under Outbound Rules, using the same IP address in the Scope tab. Name it Block IP 123.45.67.89 - Outbound.

4. **Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.**

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Key Difference	Hardware Firewall	Software Firewall
Location	**A physical device installed between a network and the internet.**	**A program installed on an individual computer or server.**
Protection	**Protects multiple devices across an entire network.**	**Primarily protects the individual device on which it runs.**
Cost and Management	**Usually more expensive and requires dedicated setup and maintenance.**	**Often included with the operating system, making it more affordable and easier to manage.**

Use Cases

1. Hardware Firewall: A large office with 200 employees uses a hardware firewall to monitor and filter incoming and outgoing traffic for the entire company network. It helps prevent unauthorized access and protect multiple computers at once.

2. Software Firewall: A student uses Windows Defender Firewall on a personal laptop to block suspicious connections and control which applications can communicate over the network.